



Name of the applicant

Annika Richter

Applicants must have a current MPIB contract or stipend; affiliation with the MPIB is not sufficient. Please note that each department has different procedures for submitting ethics applications. Ensure that the relevant people in your department have reviewed this application prior to submission (e.g., project leader, PhD supervisor, etc.).
The ethics committee reserves the right to refuse to review studies that we are not qualified to evaluate (e.g., MRI, genetic or animal studies).

Name of participating researchers

Annika Richter, Jona Carmon, Omar Sherif, Simone Kühn

Name of the research group / center and project name if applicable (e.g., for LIP, grants, etc.)

CEN

Name of the planned study

Social Interaction Experience

Planned start of data collection (enter a date)

28.07.25

Short description of the study (400 words maximum). Include information about the motivation for the study, planned methods, and the expected knowledge gain.

This study investigates the psychological experience and dynamics of text-based conversations with AI chatbots compared to human partners, with a particular focus on the experience of flow - a state of deep engagement (Csikszentmihalyi, 1990) - as well as perceived social presence (Kreijns et al., 2022; Short et al., 1976).

Participants will engage in prompted social interactions either with other humans or an AI-chatbot via a chat interface (prompts and background provided in PromptAttachement). After each chat, participants complete brief self-report measures on an interleave page, rating their experience of flow and the perceived social presence of their conversation partner in likert-scale questions. In addition the chats are analyzed for behavioral indicators, including message length, timing, message type, and conversational structure, offering insights into the underlying social dynamics of the interaction. As physiological measurements, heart-rate variability and skin conductance are used as physiological markers for arousal and measured continuously throughout the experiment (Tozman et al., 2015). As flow has been associated with an inverted U-shape heart-rate variability, the physiological measurements provide a measurement for momentary access to optimal experience during the chats (Ekman et al., 2012; Norsworthy et al., 2021; Tozman et al., 2015).

Participants are invited to the lab and measured in parallel dyads. There are two chatting conditions: human-human and human-AI. In the human-human condition, participants interact with the other participant, who is measured simultaneously. In the human-AI with the AI Chatbot, based on OpenAI 's GPT-4o included via direct API call (OpenAI, 2024). We plan to recruit 100 participants.

By integrating subjective experience, behavioral patterns, and physiological responses, this research provides a rich, multimodal understanding of chat-based social interaction. The expected outcome is a comprehensive dataset that advances our understanding of how AI-mediated conversations can support engaging and human-centric experiences. These insights aim to inform the design of future conversational AI systems that are not only functionally effective but also socially

Research instruments



Standard research instruments

Please list your standard instruments with citation either in the short description of your study above or in an additional appendix.

Specially developed research instruments

Please submit an additional document appendix (PDF, usually 1–2 pages) describing any non-standard aspects of the study, such as a particular recruiting procedure, experimental paradigms, handling of participants (e.g., in infant studies). Feel free to use figures to make this description as clear as possible.

How are research participants recruited (e.g., through the Internet, from an existing database, or via an organization)? Please provide a brief description of your recruitment procedure.

We will recruit participants via Castellum. The study will be conducted in person at the institute using an internally developed setup that incorporates APIs, including ChatGPT.

Further information on participants. Please provide number, age range and any special selection criteria.

Number of participants, age range, special selection criteria (e.g., social background, ethnicity, diseases, disorder, etc.)

We will recruit a total of 100 participants for the initial study. All participants will be over 18 years of age.

Individuals who have taken antidepressants or beta blockers during the two weeks prior to the experiment will be excluded, as these medications can affect heart rate variability.

Does the study involve minor participants (or other participants incapable of autonomous consent)?

☐ Yes

Please add the Declaration of Consent for Minors to your application (as well as the Declaration of Consent for Adult Participants if applicable to some).

Note that any research staff who will be in contact with children as part of the study must have proof of a background check (erweitertes polizeiliches Führungszeugnis) that was issued no earlier than five years before the planned conclusion of the study. Please be aware that individual schools or KitAs may require this certificate to be issued for other time periods (e.g., the last six months).

Please explain how the study procedure will guarantee informed consent and protect such participants:

☒ No

Please add the Declaration of Consent for Adult Participants to your application.

The research involves

☐

Mood induction

☐

Hidden observation

☐

Deception of the participant

Subjects must be thoroughly debriefed and informed about the purpose of the research project following the assessment.

Justify why this manipulation is necessary and provide a written description of your debriefing procedure:

☐

Physical risks

☐

Mental risks

The participants (or their legal guardians) must be informed about any risks before the study begins and confirm by signature that they agree to the conditions of the study.

Justify why this is necessary and explain how the risk of negative consequences is minimized:

☐

Other ethically sensitive procedure

Please specify and justify why this ethically sensitive procedure is necessary:

☒

None of them

Are participants asked to divulge any sensitive information?

Information is sensitive if:

(a) it could, were it to become known outside of the research project, negatively affect the participant's reputation, compromise the participant's employment situation, lead to complications with the police, or otherwise negatively affect the participant.

(b) it could potentially allow to identify participants, for instance in combination with other collected or freely available information.

Participants must be informed that such information will be requested before the study begins, and confirm by signature that they agree to the conditions of the study.

- | | | |
|---|--|--|
| ☐ Race or ethnicity | ☐ Union membership | ☐ Sexual behavior |
| ☐ Political opinion | ☐ Health | ☐ Drug abuse |
| ☐ Religious or philosophical beliefs | ☐ Genetic or biometric data | ☐ Financial situation |
| | | ☒ Other |

Please specify the information and explain why it is necessary:

Participants engage in prompted social interactions that unfold in an open-ended manner. As such, they may voluntarily disclose sensitive information. It is neither possible nor appropriate to fully control the behavior of the AI chatbot or the human conversation partner; conversations are intended to evolve organically. Consequently, there is a risk that discussions may include sensitive or emotionally challenging content. While the study is designed to foster engaging and immersive experiences, these interactions may also elicit negative emotions.

This potential risk is proactively addressed in the participant materials. Participants are explicitly informed of the possibility of sensitive content and are instructed to avoid sharing personally identifying or emotionally charged information. A code of conduct promotes respectful communication, and participants are clearly reminded that they may withdraw from the study at any time without penalty. These safeguards are detailed in the study materials and form an integral part of the informed consent process.

☐ No sensitive information will be collected

Will participants be remunerated for their participation?

☒ Yes

Participants must be informed of the amount of remuneration to be received before the study begins. If applicable, participants must also be informed that not everyone will be paid the same amount in this study and why this is the case.

How will the anonymity of the participants be ensured? e.g. bank details and receipts stored separately from experimental data and locked up

In line with the institute's compensation policy, participants will receive a base participation fee of €13 per hour. The expected duration of the study is approximately 60 minutes. The final compensation may be adjusted based on the median duration of actual participation across all sessions, with a guaranteed minimum rate of €13/hour for all participants. In addition, participants will receive a €5 bonus for attending the session in person at the institute.

Compensation will be issued in cash after the participant has completed the study. No personally identifiable information (e.g., name, signature, or bank account details) will be collected or stored.

Participant identification information retrieved via Castellum will be stored only until the end of the data collection phase. This information is used solely to prevent multiple enrollments by the same individual and will be deleted once recruitment is complete.

The consent form will be presented within the web application. Participants provide consent by ticking the corresponding checkboxes (e.g., "I consent") for the relevant items. It is not possible to proceed with the experiment unless all required consent boxes have been checked, which ensures that participation is based on informed consent.

No personal identifiers (e.g., names, signatures, or contact details) will be stored within the consent form or linked to experimental data. This procedure guarantees that participation remains fully anonymous.

What provision has been made for the remuneration of participants who drop out of the study prematurely?

Participants who withdraw from the study prematurely will receive partial compensation proportionate to the time they have participated in the experiment, including the bonus for attending the session in person. The exact amount will be calculated based on the time and effort already invested up to the point of withdrawal. This approach ensures fair treatment and acknowledges their contribution, even if they do not complete the entire study.

☐ No

Data will be stored in the following format (check all that apply):

☐ Paper

☒ Electronic

Will data be collected in the format of audio or video recordings?

☐ Yes

Please make sure to inform participants about such recordings in the consent form. If you plan to use video/audio recordings for purposes other than strict data analysis (i.e. presentation, publication etc.), please include the separate consent form for audio and video recordings in your application.

☒ No

How will research data be handled on a daily basis to ensure that participants cannot be identified?

All personal data (e.g., consent forms, bank details, receipts, contact information etc.) need to be stored separately from experimental data. If you use a key to link experimental data to individuals: ensure that the key is stored separately and password protected from experimental data and access is limited to a single involved researcher or very few people. If you collect data that can be used to identify individuals: ensure that this data is stored locked up (e.g., completed questionnaires, test book-lets) or password protected (e.g., audiovisual recordings), and access is limited to a single involved researcher or very few people.

- ☐ 1. Random ID, no key linking ID to individual, no data that can be used to identify individuals collected
- ☐ 2. Random ID, no key linking ID to individual, data that can be used to identify individuals collected
- ☒ 3. Random ID, key linking ID to individual, no data that can be used to identify individuals collected
- ☐ 4. Random ID, key linking ID to individual, data that can be used to identify individuals collected
- ☐ 5. Other

Please specify:

Only Annika Richter, Jona Carmon, and Simone Kühn will have access to the raw data files. Participation verification for compensation is handled independently and cannot be linked to the experimental data. Consequently, any data shared at later stages (e.g., during peer review or publication) will not include any information that could identify or link experimental data to individual participants.

If you keep a key linking experimental data to individuals, when will this key be deleted?

e.g., after publication of the study, after 5 years, ...

The unique participant keys provided by Castellum will be securely deleted immediately after the completion of data collection. These keys are used solely to ensure that no individual participates in the study more than once. Once data collection is complete, the keys will be removed to guarantee that no connection can be made between participants' identities and the data collected during the experiment.

Will data be made accessible to other researchers or the public (e.g. through an online repository or archive, such as the Open Science Framework, online supplemental material of the publication, or other websites)?

☒ Yes

Please make sure that the consent form describes correctly how you will be sharing the experimental data of your study in the future. NOTE: You can only share experimental data that does not allow identification of participants. If other types of data must be shared, please contact the MPI data protection representative and provide a pdf of data sharing procedures.

☐ No

Will data be collected, stored, or processed in full or in part by an external contractor?

☒ Yes

Please refer to the Max Planck Society's OHB Chapter XVII.03 and check with the MPI data protection representative whether using the agreement for commissioned data processing) is advisable.

☐ No

General information regarding data protection regulations.

By checking the boxes below, you are confirming that you will adhere to the data protection requirements.

- ☒ Any personal data must be protected from unauthorized access. "Physical" data (e.g., consent forms, audiovisual data) must be kept locked up. "Electronic" data must be protected by passwords.
- ☒ Research data used on a daily basis must not contain any reference to names or other personal information to ensure that participants cannot be identified.
- ☒ Potential participants may only be added to the database with their consent. If participants request to be removed from the database, their data must be deleted immediately and permanently.
- ☒ Access to any personal data or data that can be used to link study data to individuals is limited to a single person or very few people. Any person who can access these data must conform to the data protection regulations.

Family name

Richter

Given name

Annika

Signature

mpib.protect Jamf Protect CSR Identity
Digitally signed by mpib.protect Jamf Protect CSR Identity
Date: 2025.07.01 18:41:26 +02'00'

Date

25/06/2025

I certify that the information provided above is accurate and complete to the best of my knowledge.

Date

25/06/2025

Signature

mpib.protect Jamf Protect CSR Identity
Digitally signed by mpib.protect Jamf Protect CSR Identity
Date: 2025.07.01 18:41:37 +02'00'

Reset form